Sub Queries

1.Create a report that displays EMPNO, ENAME, DEPTNO of employees who work with that employee. (Nested Sub Query)

Query:

SELECT employee_id, first_name, department_id
FROM employees
WHERE job_id=(SELECT job_id FROM employees WHERE first_name='Steven');

Output:

```
SQL> SELECT employee_id, first_name, department_id
  2  FROM employees
  3  WHERE job_id=(SELECT job_id FROM employees WHERE first_name='Steven');

EMPLOYEE_ID FIRST_NAME                                           DEPARTMENT_ID
----------- -------------------------------------------------- -------------
        100 Steven                                                        90

SQL>
```

2.Display the employees who earn more than the average salary of EMP table. (Nested Sub Query)

Query:

SELECT employee_id,first_name,salary
FROM employees
WHERE salary>ALL(SELECT AVG(salary) FROM employees GROUP BY department_id);

Output:

```
     EMPNO ENAME             SAL
---------- ---------- ----------
      7566 JONES            2975
      7698 BLAKE            2850
      7782 CLARK            2450
      7788 SCOTT            3000
      7839 KING             5000
      7902 FORD             3000

6 rows selected.
```

3.Display the ENAME, JOB who are Managers. (EXISTS Operator)

Query:

SELECT employee_id,first_name,job_id
FROM employees
WHERE employee_id IN (SELECT manager_id FROM departments);

Output:

```
SQL> SELECT employee_id,first_name,job_id
  2  FROM employees
  3  WHERE employee_id IN (SELECT manager_id FROM departments);

EMPLOYEE_ID FIRST_NAME
----------- --------------------------------------------------
JOB_ID
--------------------
        200 Jennifer
AD_ASST

        201 Michael
MK_MAN

        114 Den
PU_MAN


EMPLOYEE_ID FIRST_NAME
----------- --------------------------------------------------
JOB_ID
--------------------
        203 Susan
HR_REP

        121 Adam
ST_MAN

        103 Alexander
IT_PROG


EMPLOYEE_ID FIRST_NAME
----------- --------------------------------------------------
JOB_ID
--------------------
        100 Steven
AD_PRES


7 rows selected.

SQL> |
```

4.Display the employees who earn less than the least salary of DEPTNO 10. (ALL Operator)

Query:

SELECT first_name,last_name,salary,department_id
FROM employees
WHERE salary=(SELECT MIN(salary) FROM employees WHERE department_id=10);

Output:

```
SQL> SELECT first_name,last_name,salary,department_id
  2  FROM employees
  3  WHERE salary=(SELECT MIN(salary) FROM employees WHERE department_id=10);

FIRST_NAME
--------------------------------------------------
LAST_NAME                                              SALARY DEPARTMENT_ID
-------------------------------------------------- ---------- -------------
Jennifer
Whalen                                                   4400            10


SQL> |
```

5. Display the employees who have the same DEPTNO and MGR of a given employee, excluding that employee. (Nested Sub Query)

Query:

SELECT empno, ename, deptno, mgr

FROM emp

WHERE (deptno, mgr) =(SELECT deptno, mgr

FROM emp

WHERE ename=’Ward'

)AND ename<>'Ward';

Output:

```
SQL> SELECT empno, ename, deptno, mgr
  2  FROM emp
  3  WHERE (deptno, mgr) =
  4  (
  5      SELECT deptno, mgr
  6      FROM emp
  7      WHERE ename = 'WARD'
  8  )
  9  AND ename <> 'WARD';

     EMPNO ENAME          DEPTNO        MGR
---------- ---------- ---------- ----------
      7499 ALLEN              30       7698
      7654 MARTIN             30       7698
      7844 TURNER             30       7698
      7900 JAMES              30       7698

SQL> |
```

6.Display the employee number and name of all employees who work in a department with any employee whose name contains 'R'. (Nested Sub Query)

Query:

SELECT employee_id,first_name||' '||last_name AS name
FROM employees

WHERE department_id IN (SELECT department_id FROM employees WHERE LOWER(first_name||last_name) LIKE '%r%');

Output:

```
SQL> SELECT employee_id,first_name||' '||last_name AS name
  2  FROM employees
  3  WHERE department_id IN (SELECT department_id FROM employees WHERE LOWER(first_name||last_name) LIKE '%r%');

EMPLOYEE_ID
-----------
NAME
--------------------------------------------------------------------------------
        100
Steven King

        101
Neena Kochhar

        102
Lex De Haan


EMPLOYEE_ID
-----------
NAME
--------------------------------------------------------------------------------
        103
Alexander Hunold

        104
Bruce Ernst

        105
David Austin


EMPLOYEE_ID
-----------
NAME
--------------------------------------------------------------------------------
        106
Valli Pataballa

        107
Diana Lorentz

        108
Nancy Greenberg


EMPLOYEE_ID
-----------
NAME
--------------------------------------------------------------------------------
        109
Daniel Faviet

        110
John Chen

        111
Ismael Sciarra


EMPLOYEE_ID
-----------
```

7. Display the ENAME, DEPTNO, JOB of all employees who work in NEW YORK. (Nested Sub Query)

Query:

SELECT employee_id,first_name,department_id
FROM employees
WHERE department_id IN (SELECT department_id FROM departments WHERE location_id=(SELECT location_id FROM locations WHERE city='South San Francisco'));

Output:

```
SQL> SELECT employee_id,first_name,department_id
  2  FROM employees
  3  WHERE department_id IN (SELECT department_id FROM departments WHERE location_id=(SELECT location_id FROM locations WHERE city='South San Francisco'));

EMPLOYEE_ID FIRST_NAME                                         DEPARTMENT_ID
----------- -------------------------------------------------- -------------
        120 Matthew                                                       50
        121 Adam                                                          50
        122 Payam                                                         50
        123 Shanta                                                        50
        124 Kevin                                                         50
        176 Jonathon                                                      50
        177 Jack                                                          50
        178 Kimberely                                                     50

8 rows selected.

SQL> |
```

8. Modify the query so that the user is prompted for a LOC.

Query:
SELECT ENAME, DEPTNO, JOB FROM EMP WHERE DEPTNO=(SELECT DEPTNO FROM DEPT WHERE LOC='&LOC');

Output:

```
SQL> SELECT ename, deptno, job
  2  FROM emp
  3  WHERE deptno =
  4  (
  5      SELECT deptno
  6      FROM dept
  7      WHERE loc='&LOC'
  8  );
Enter value for loc: Dallas
old   7:     WHERE loc='&LOC'
new   7:     WHERE loc='Dallas'

no rows selected

SQL> |
```

9. Display the name and salary of every employee who reports to KING.

Query:
SELECT first_name,salary
FROM employees
WHERE manager_id=(SELECT employee_id FROM employees WHERE last_name='King');

Output:

```
SQL> SELECT first_name,salary
  2  FROM employees
  3  WHERE manager_id=(SELECT employee_id FROM employees WHERE last_name='King');

FIRST_NAME                                               SALARY
-------------------------------------------------------- ----------
Neena                                                         17000
Lex                                                           17000
Den                                                           11000
Matthew                                                        8000
Adam                                                           8200
Payam                                                          7900
Shanta                                                         6500
Kevin                                                          5800
Michael                                                       13000

9 rows selected.

SQL> |
```

10. Display all the employees working with JAMES.

Query:

SELECT * FROM employees
WHERE department_id=(SELECT department_id FROM employees WHERE first_name='James');

Output:

```
SQL> SELECT * FROM employees
  2  WHERE department_id=(SELECT department_id FROM employees WHERE first_name='James');

EMPLOYEE_ID FIRST_NAME
----------- ---------------------------------------------
LAST_NAME
---------------------------------------------
EMAIL                                         PHONE_NUMBER
--------------------------------------------- --------------------
HIRE_DATE JOB_ID                  SALARY MANAGER_ID DEPARTMENT_ID
--------- -------------------- ---------- ---------- -------------
        203 Susan
Mavris
SMAVRIS                                       111133
07-JUN-02 HR_REP                    6500        101            40

EMPLOYEE_ID FIRST_NAME
----------- ---------------------------------------------
LAST_NAME
---------------------------------------------
EMAIL                                         PHONE_NUMBER
--------------------------------------------- --------------------
HIRE_DATE JOB_ID                  SALARY MANAGER_ID DEPARTMENT_ID
--------- -------------------- ---------- ---------- -------------
        204 James
Smith
JSMITH                                        111134
21-SEP-05 HR_REP                    6000        203            40


SQL> |
```

11**. Display all the employees who earn less than the average salary of their respective departments. (Correlated Sub Query)**

Query:

SELECT first_name,salary,department_id
FROM employees e
WHERE salary<(SELECT AVG(salary) FROM employees WHERE department_id=e.department_id);

Output:

```
SQL> SELECT first_name,salary,department_id
  2  FROM employees e
  3  WHERE salary<(SELECT AVG(salary) FROM employees WHERE department_id=e.department_id);

FIRST_NAME                                          SALARY DEPARTMENT_ID
-------------------------------------------------- ---------- -------------
Neena                                                17000            90
Lex                                                  17000            90
David                                                 4800            60
Valli                                                 4800            60
Diana                                                 4200            60
Daniel                                                9000            20
John                                                  8200            20
Ismael                                                7700            20
Jose                                                  7800            20
Luis                                                  6900            20
Alexander                                             3100            30

FIRST_NAME                                          SALARY DEPARTMENT_ID
-------------------------------------------------- ---------- -------------
Shelli                                                2900            30
Sigal                                                 2800            30
Guy                                                   2600            30
Karen                                                 2500            30
Shanta                                                6500            50
Kevin                                                 5800            50
Kimberely                                             7000            50
James                                                 6000            40

19 rows selected.

SQL> |
```

12. Display the LOC and average salary of each location. (Scalar Sub Query)

Query:
SELECT d.loc,(

SELECT AVG(e.sal)

FROM emp e

WHERE e.deptno=d.deptno) AS avg_salary FROM dept d;

Output:

```
SQL> SELECT d.loc,
  2  (
  3      SELECT AVG(e.sal)
  4      FROM emp e
  5      WHERE e.deptno=d.deptno
  6  ) AS avg_salary
  7  FROM dept d;

LOC           AVG_SALARY
------------- ----------
DALLAS              2175
CHICAGO       1566.66667
NEW YORK      2916.66667
BOSTON

SQL> |
```

13. Display the least N salaries. (Inline View)

Query:
SELECT * FROM (SELECT employee_id,first_name,salary FROM employees ORDER BY salary) WHERE ROWNUM<=5;

Output:

```
SQL> SELECT * FROM (SELECT employee_id,first_name,salary FROM employees ORDER BY salary) WHERE ROWNUM<=5;

EMPLOYEE_ID FIRST_NAME                                           SALARY
----------- ---------------------------------------------------- ----------
        119 Karen                                                      2500
        118 Guy                                                        2600
        117 Sigal                                                      2800
        116 Shelli                                                     2900
        115 Alexander                                                  3100

SQL> |
```

14. Display the last N rows from the EMP table. (Correlated Sub Query)

Query:

SELECT * FROM (SELECT * FROM employees ORDER BY employee_id DESC) WHERE ROWNUM<=5;

Output:

```
        115 Alexander                                                  3100

SQL> SELECT * FROM (SELECT * FROM employees ORDER BY employee_id DESC) WHERE ROWNUM<=5;

EMPLOYEE_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL                                              PHONE_NUMBER
-------------------------------------------------- --------------------
HIRE_DATE JOB_ID                   SALARY MANAGER_ID DEPARTMENT_ID
--------- -------------------- ---------- ---------- -------------
        204 James
Smith
JSMITH                                             111134
21-SEP-05 HR_REP                     6000        203            40

EMPLOYEE_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL                                              PHONE_NUMBER
-------------------------------------------------- --------------------
HIRE_DATE JOB_ID                   SALARY MANAGER_ID DEPARTMENT_ID
--------- -------------------- ---------- ---------- -------------
        203 Susan
Mavris
SMAVRIS                                            111133
07-JUN-02 HR_REP                     6500        101            40

EMPLOYEE_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL                                              PHONE_NUMBER
-------------------------------------------------- --------------------
HIRE_DATE JOB_ID                   SALARY MANAGER_ID DEPARTMENT_ID
--------- -------------------- ---------- ---------- -------------
        201 Michael
Hartstein
MHARTSTE                                           111132
17-FEB-04 MK_MAN                    13000        100            20

EMPLOYEE_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL                                              PHONE_NUMBER
-------------------------------------------------- --------------------
HIRE_DATE JOB_ID                   SALARY MANAGER_ID DEPARTMENT_ID
--------- -------------------- ---------- ---------- -------------
        200 Jennifer
Whalen
JWHALEN                                            111131
17-SEP-03 AD_ASST                    4400        101            10

EMPLOYEE_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL                                              PHONE_NUMBER
-------------------------------------------------- --------------------
HIRE_DATE JOB_ID                   SALARY MANAGER_ID DEPARTMENT_ID
--------- -------------------- ---------- ---------- -------------
        178 Kimberely
Grant
KGRANT                                             111130
24-MAY-07 SA_REP                     7000        120            50

SQL> |
```

15. Display the employees and sort only employees working in DALLAS. (Scalar Sub Query)

Query:

SELECT * FROM employees WHERE department_id IN (SELECT department_id FROM departments WHERE location_id=(SELECT location_id FROM locations WHERE city='Dallas')) ORDER BY salary;

Output:

```
12 rows selected.

SQL> SELECT * FROM employees WHERE department_id IN (SELECT department_id FROM departments WHERE location_id=(SELECT location_id FROM locations WHERE city='Dallas')) ORDER BY salary;

EMPLOYEE_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL                                              PHONE_NUMBER
-------------------------------------------------- --------------------
HIRE_DATE JOB_ID                   SALARY MANAGER_ID DEPARTMENT_ID
--------- -------------------- ---------- ---------- -------------
        107 Diana
Lorentz
DLORENTZ                                           888888
07-FEB-07 IT_PROG                    4200        103            60

EMPLOYEE_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL                                              PHONE_NUMBER
-------------------------------------------------- --------------------
HIRE_DATE JOB_ID                   SALARY MANAGER_ID DEPARTMENT_ID
--------- -------------------- ---------- ---------- -------------
        105 David
Austin
DAUSTIN                                            666666
25-JUN-05 IT_PROG                    4800        103            60

EMPLOYEE_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL                                              PHONE_NUMBER
-------------------------------------------------- --------------------
HIRE_DATE JOB_ID                   SALARY MANAGER_ID DEPARTMENT_ID
--------- -------------------- ---------- ---------- -------------
        106 Valli
Pataballa
VPATABAL                                           777777
05-FEB-06 IT_PROG                    4800        103            60

EMPLOYEE_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL                                              PHONE_NUMBER
-------------------------------------------------- --------------------
HIRE_DATE JOB_ID                   SALARY MANAGER_ID DEPARTMENT_ID
--------- -------------------- ---------- ---------- -------------
        104 Bruce
Ernst
BERNST                                             555555
21-MAY-07 IT_PROG                    6000        103            60

EMPLOYEE_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL                                              PHONE_NUMBER
-------------------------------------------------- --------------------
HIRE_DATE JOB_ID                   SALARY MANAGER_ID DEPARTMENT_ID
--------- -------------------- ---------- ---------- -------------
        113 Luis
Popp
LPOPP                                              111116
07-DEC-07 FI_ACCOUNT                 6900        108            20

EMPLOYEE_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL                                              PHONE_NUMBER
-------------------------------------------------- --------------------
HIRE_DATE JOB_ID                   SALARY MANAGER_ID DEPARTMENT_ID
--------- -------------------- ---------- ---------- -------------
        111 Ismael
Sciarra
ISCIARRA                                           111114
30-SEP-05 FI_ACCOUNT                 7700        108            20

EMPLOYEE_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL                                              PHONE_NUMBER
-------------------------------------------------- --------------------
HIRE_DATE JOB_ID                   SALARY MANAGER_ID DEPARTMENT_ID
--------- -------------------- ---------- ---------- -------------
        112 Jose
Manuel
JMANUEL                                            111115
07-MAR-06 FI_ACCOUNT                 7800        108            20

EMPLOYEE_ID FIRST_NAME
```

16. Display the employees who earn a salary less than the average salary of their respective department. Also display the average salary. (Inline View)

```
12 rows selected.

SQL> SELECT e.empno,
            e.ename,
            e.sal,
            v.avg_sal
     FROM emp e
     JOIN
     (
         SELECT deptno,
                AVG(sal) avg_sal
         FROM emp
         GROUP BY deptno
     ) v
     ON e.deptno=v.deptno
     WHERE e.sal<v.avg_sal;

     EMPNO ENAME             SAL    AVG_SAL
---------- ---------- ---------- ----------
      7369 SMITH             800       2175
      7876 ADAMS            1100       2175
      7521 WARD             1250 1566.66667
      7654 MARTIN           1250 1566.66667
      7844 TURNER           1500 1566.66667
      7900 JAMES             950 1566.66667
      7782 CLARK            2450 2916.66667
      7934 MILLER           1300 2916.66667

8 rows selected.

SQL>
```

17. Display the LOC of those departments whose total salary is less than the average salary of all employees. (WITH Clause)

Query:
WITH dept_sal AS (
SELECT d.location_id,SUM(e.salary) total_sal FROM departments d JOIN employees e ON d.department_id=e.department_id GROUP BY d.location_id)
SELECT * FROM dept_sal;

Output:

```
SQL> WITH dept_sal AS (
  SELECT d.location_id,SUM(e.salary) total_sal FROM departments d JOIN employees e ON d.department_id=e.department_id GROUP BY d.location_id)
  SELECT * FROM dept_sal;

LOCATION_ID  TOTAL_SAL
----------- ----------
       1100      74900
       1200      93400
       1300      24900
       1000      60400

SQL>
```